A Comparative Study of CNN-LSTM Architectures for Visual Question Answering

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***Abstract*—**

***Index Terms*—**

I. Introduction

II. Related Work

Visual Question Answering (VQA) sits at the intersection of computer vision and natural language processing, requiring a system to jointly understand an image and a textual question to produce a correct answer. Research in this space has driven advances in visual encoding, sequence modeling, cross-modal fusion, and answer generation. This section reviews the key prior works underlying the components of our proposed system.

Convolutional Neural Networks (CNNs) are the standard backbone for visual feature extraction in VQA systems. He et al. [1] proposed residual learning, which reformulates each layer as learning a residual function with respect to its input rather than an unreferenced mapping. This design allows networks to be trained to much greater depth without degradation, and ResNet variants consistently achieve strong performance across visual recognition benchmarks. We adopt pretrained ResNet-18 and ResNet-50 as our image encoders, retaining spatial feature maps from the final convolutional layer rather than the global pooled vector, so that positional information across image regions is preserved for the subsequent attention mechanism.

TABLE I
DATASET STATISTICS AND SPLITS OF VIINFOGRAPHICVQA.

Subset	Images	Groups	QAs
Single-Image (train)	1788	–	12521
Single-Image (test)	192	–	1374
Multi-Image (train)	4823	2090	5878
Multi-Image (test)	509	218	636
Total train	6096	2090	18399
Total test	651	218	2010

III. Model Architecture

IV. Experiments

V. Conclusion

References

[1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *CoRR*, vol. abs/1512.03385, 2015. [Online]. Available: http://arxiv.org/abs/1512.03385

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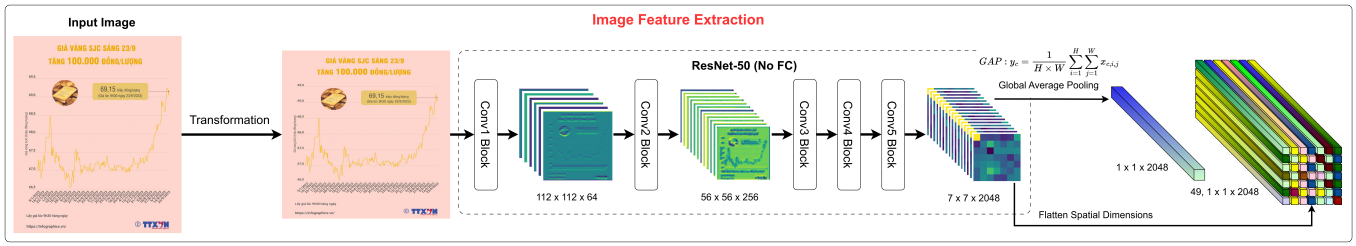

Fig. 1. CNN feature extraction pipeline. An input infographic is encoded by ResNet-50 into a spatial feature tensor and then reshaped into regional vectors for downstream attention.